

Normal distribution

Thursday, March 7, 2024 10:10 AM

Example 59. Given standard normal distribution, find the values of k such that

(a) $P(Z > k) = 0.3015$

(b) $P(k < Z < -0.18) = 0.4197$

a) Given $P(Z > k) = 0.3015$

$$P(0 < Z < \infty) = 0.5$$

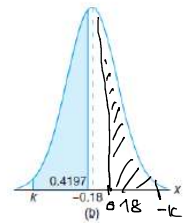
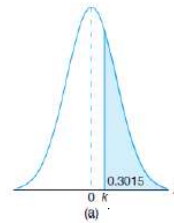
$$P(0 < Z < k) + P(k < Z < \infty) = 0.5$$

$$P(0 < Z < k) = 0.5 - P(Z > k)$$

$$= 0.5 - 0.3015$$

$$P(0 < Z < k) = 0.2015$$

$$k = 0.53$$



$$P(0 < Z < k)$$

b) $P(k < Z < -0.18) = 0.4197$

$$P(k < Z < -0.18) = P(0.18 < Z < -k)$$

$$0.4197 = P(0 < Z < -k) - P(0 < Z < 0.18)$$

$$0.4197 = P(0 < Z < -k) - 0.0714$$

$$P(0 < Z < -k) = 0.4197 + 0.0714$$

$$P(0 < Z < -k) = 0.4911$$

$$-k = 2.37 \Rightarrow k = -2.37$$

$$\int e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \sim \int e^{-\frac{1}{2}z^2} dz, \quad z = \frac{x-\mu}{\sigma}$$

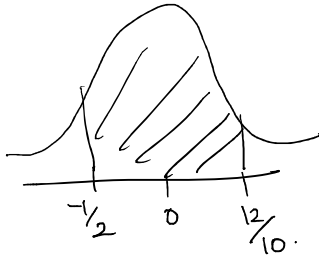
$$\underline{\underline{N(\mu, \sigma)}} \sim \underline{\underline{N(0, 1)}}$$

Example 60. Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$. Find the probability that X assumes a value between 45 and 62.

$$P(45 < X < 62) = P\left(\frac{45-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{62-\mu}{\sigma}\right) \sim N(\mu, \sigma)$$

$$= P\left(\frac{45-50}{10} < Z < \frac{62-50}{10}\right)$$





$$\begin{aligned}
 &= P\left(\frac{45-50}{10} < z < \frac{62-50}{10}\right) \\
 &= P\left(-\frac{1}{2} < z < \frac{12}{10}\right) \sim N(0,1) \\
 &= P\left(-\frac{1}{2} < z < 0\right) + P\left(0 < z < \frac{12}{10}\right) \\
 &= P\left(0 < z < \frac{1}{2}\right) + P\left(0 < z < \frac{12}{10}\right) \\
 &= 0.1915 + 0.3849.
 \end{aligned}$$

Example 61. The weekly wages of 1000 workers are normally distributed around a mean of Rs.70 with standard deviation of Rs.5. Estimate the number of workers whose weekly wages will be

- between Rs.69 and Rs.72
- less than Rs.69
- more than Rs.72

Given $\mu = 70, \sigma = 5$.

$$\begin{aligned}
 P(69 < x < 72) &= P\left(\frac{69-70}{5} < z < \frac{72-70}{5}\right) \quad \left. \begin{array}{l} \text{for 1000 workers} \\ = 1000 \times P(69 < x < 72) \\ = 234.7 \end{array} \right\} \\
 &= 0.2347. \\
 P(x < 69) &= 0.079 \quad \left. \begin{array}{l} \text{for 1000 workers} \\ = 1000 \times P(x < 69) \\ = 79. \end{array} \right\} \\
 P(x > 72) &= 0.8447.
 \end{aligned}$$

Definition 28. Gamma function $\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$

Definition 29. Gamma distribution The continuous random variable X has a gamma distribution, with parameters α and β , if its density function is $f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} & x > 0 \\ 0 & \text{otherwise} \end{cases}$

Definition 30. The special gamma distribution for which $\alpha = 1$ is called the exponential distribution.

$$\int e^{-x} dx \quad \int x e^{-x} \quad \int x^4 e^{-x} \quad \int x^{100}$$

① $f(x) \geq 0$.

$$\begin{aligned}
 \textcircled{2} \quad \int_0^{\infty} f(x) dx &= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} x^{\alpha-1} e^{-x/\beta} dx \quad \frac{x}{\beta} = t \\
 &= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} \beta^\alpha t^{\alpha-1} e^{-t} dt
 \end{aligned}$$

$$\frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1$$

Multiple regression

Monday, March 18, 2024 11:33 AM

x	y	z
3	2	4
2	1	8
5	4	3
6	2	2

Find the multiple regression equation.

$$z = ax + by + c \Rightarrow \sum z = \sum ax + \sum by + \sum c$$

$$xz = ax^2 + bxy + cx \Rightarrow \sum xz = a\sum x^2 + b\sum xy + c\sum x$$

$$yz = axy + by^2 + cy \Rightarrow \sum yz = a\sum xy + b\sum y^2 + c\sum y$$

x	y	z	x ²	y ²	xy	xz	yz
3	2	4	9	4	6	12	8
2	1	8	4	1	2	16	8
5	4	3	25	16	20	15	12
6	2	2	36	4	12	12	4

$$17 = 16a + 9b + 4c$$

$$55 =$$

Testing of hypothesis

Monday, March 11, 2024 9:22 PM

Example 73. • Average daily wage of workers in vellore > or < national Average

- Given national average = Rs 200

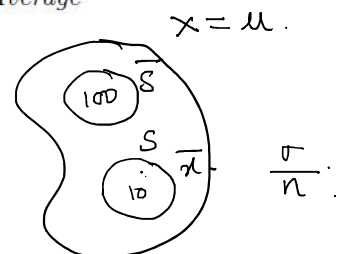
Sample mean = $\bar{x} = 199$, 200

Population mean = $\mu = 200$

$$H_0: \bar{x} = \mu$$

$$H_1: \bar{x} \neq \mu$$

$$Z = \frac{\bar{x} - \mu}{SE}, \text{ standard error: SE}$$



Example 75. Test for significance for single proportion:

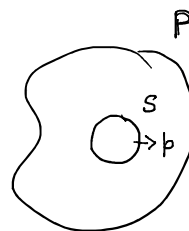
- p = sample proportion

- P = Population proportion

- Test statistic: $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$, $P + Q = 1$

- Limits for population proportion P is $p \pm 3\sqrt{\frac{pq}{n}}$

- 98 percent confidence limits for population proportion are $p \pm 2.33\sqrt{\frac{pq}{n}}$



Example 76. In a city, a sample of 1000 people were taken and out of them 540 are veg and rest are non veg. Can we say that both habits of eating are equally popular in the city at 1 percent level of significance.

$$P = \frac{1}{2}, \text{ sample prop. } p.$$

$$H_0: p = \frac{1}{2}, H_1: p \neq \frac{1}{2}$$

Sample given: $n = 1000$, sample prop $p = \frac{540}{1000}$

$$Z = \frac{p - P}{S.E}, S.E = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{\frac{1}{2} \times \frac{1}{2}}{1000}}$$

$$Z = \frac{\frac{540}{1000} - \frac{1}{2}}{0.015} = 2.53$$

$$Z = 2.53$$

$$Z_{1\%} = 2.58 \Rightarrow |Z| < |Z_{1\%}|$$

Accept H_0 i.e., both eating habits are equally popular.

$$Z_{5\%} = 1.96 \Rightarrow |Z| > Z_{5\%} \text{ i.e., both eating habits are}$$

$z_{5\%} = 1.19 \Rightarrow |z| > z_{5\%}$ i.e., both eating habits are not equally popular
 Reject H_0 :

13. Critical values of Z_α :

Critical values (z_α)	Level of Significance (α)			
	1%	2%	5%	10%
Two tailed test	$ z_\alpha = 2.58$	$ z_\alpha = 2.33$	$ z_\alpha = 1.96$	$ z_\alpha = 1.645$
Right tailed test	$z_\alpha = 2.33$	$z_\alpha = 2.054$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left tailed test	$z_\alpha = -2.33$	$z_\alpha = -2.054$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

$$|z| < z_\alpha \Rightarrow \text{Accept } H_0.$$

$$\left| \frac{p - P}{S.E} \right| < z_\alpha$$

$$|x| < \alpha$$

$$-\alpha < x < \alpha$$

$$|p - P| < z_\alpha \cdot S.E$$

$$-z_\alpha \cdot S.E < p - P < z_\alpha \cdot S.E$$

Limit of confidence interval. $p - z_\alpha \cdot S.E < p < p + z_\alpha \cdot S.E$

Single sample proposition

Wednesday, March 13, 2024 10:20 PM

$H_0: p = 0.5 \rightarrow \text{manuf. loss.}$
 $H_1: p > 0.5$
 $\downarrow$ reject $\rightarrow \text{Cost}$

$p < 0.5 \rightarrow \text{manuf. -}$
 $\downarrow$ accept $\rightarrow \text{Consumer. loss}$
 $\hookrightarrow \text{health} \rightarrow \text{Cost.}$



Example 77. A die was thrown 9000 times and of these 3220 yielded a 3 or 4. Is this consistent with the hypothesis that the die was unbiased?

p - population prop -
 $= p(\text{obtaining 3 or 4}) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

p - Sample prop.
 $= \frac{3220}{9000}$

Null hyp: $H_0: p = \frac{1}{3}$, Alt. Hyp $p \neq \frac{1}{3}$

$$z = \frac{p - P}{S.E.}, S.E. = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{\frac{1}{3} \times \frac{2}{3}}{9000}}$$

$$z = \frac{\frac{3220}{9000} - \frac{1}{3}}{S.E.} = 4.4$$

$$z_\alpha = 1.96, \alpha = 5\%$$

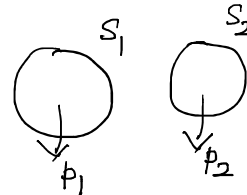
$$|z| < z_\alpha \Rightarrow \text{Accept } H_0.$$

Prob: 1



comp. S & P

pb: 2



$$\frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

$$\frac{p_1 - p_2}{S.E} \quad p = (p_1 \times p_2)$$

2. Test of Significance of Difference of Proportions:

To test the significant difference between the sample proportions p_1 and p_2 .

Null hypothesis is : $H_0 : p_1 = p_2$

Alternative hypothesis is : $H_1 : p_1 \neq p_2$

a) To test the significant difference between the sample proportions p_1 & p_2

$$\text{Test Statistic : } Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad \text{where } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

and $q = 1 - p$

b) To test the significant difference between p_1 & p

$$\text{Test Statistic : } Z = \frac{p_1 - p}{\sqrt{\frac{p(1-p)}{n_1}}} \quad \text{where } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

c) If the sample proportions are not known then $Z = \frac{p_1 - p_2}{\sqrt{\frac{p_1 Q_1}{n_1} + \frac{p_2 Q_2}{n_2}}}$

Example 78. Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same as 5 percent level.

S_1 - men

$$n_1 = 400$$

$$p_1 = \frac{200}{400}$$

S_2 - women.

$$n_2 = 600$$

$$p_2 = \frac{325}{600}$$

$$Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad , \quad p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$p = \frac{200 + 325}{1000}$$

$$Z = \frac{\frac{200}{400} - \frac{325}{600}}{\sqrt{(0.52) \left(\frac{1}{400} + \frac{1}{600} \right)}} = 1.3$$

(0.48)

$$Z_{\alpha} = 1.96 \quad \alpha = 5\%$$

$|z| < z_\alpha \Rightarrow \text{Accept } H_0.$

i.e, Men & Women are equally favorable
for the proposal.

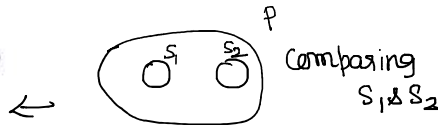
Two sample proposition

Friday, March 15, 2024 10:12 PM

a) To test the significant difference between the sample proportions p_1 & p_2

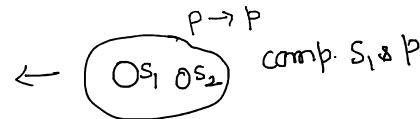
$$\text{Test Statistic : } Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad \text{where } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

and $q = 1 - p$



b) To test the significant difference between p_1 & p

$$\text{Test Statistic : } Z = \frac{p_1 - p}{\sqrt{\frac{p_1 p_1}{n_1} + \frac{p p}{n_2}}} \quad \text{where } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$



c) If the sample proportions are not known then $Z = \frac{p_1 - p_2}{\sqrt{\frac{p_1 Q_1}{n_1} + \frac{p_2 Q_2}{n_2}}}$

Example 79. In a random sample of 400 student of the university teaching department, it was found that 300 students failed in the examination. In another sample of 500 students of the affiliated colleges the number of failures in the same examination was found to be 300. Find out whether the proportion of failures in the university teaching department significantly greater than the proportion of failures in the university teaching department and affiliated colleges taken together

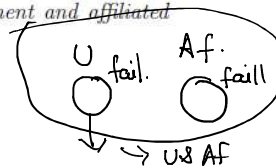
$$n_1 = 400$$

$$p_1 = \frac{300}{400}$$

$$n_2 = 500$$

$$p_2 = \frac{300}{500}$$

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{300 + 300}{900} = \frac{600}{900} = \frac{2}{3}$$



$$H_0 : p_1 = p$$

$$H_1 : p_1 > p$$

$$S.E = \sqrt{\frac{p_1 p_1}{n_1} + \frac{p p}{n_2}} = \sqrt{\frac{500 \cdot \frac{2}{3} \cdot (\frac{1}{3})}{400 \cdot \frac{2}{3} \cdot \frac{1}{3}}} = 0.02$$

$$\text{Test statistic : } z = \frac{p_1 - p}{S.E} = \frac{\frac{3}{4} - \frac{2}{3}}{0.02} = 4.7$$

$$\alpha = 5\% ; z_{\alpha} = 1.64$$

$$\Rightarrow |z| > z_{\alpha} \Rightarrow \text{Accept } H_1, \text{ Reject } H_0$$

Example 80. In an infantile paralysis epidemic 500 persons contracted the disease. 300 received no serum treatment and of them 75 became paralyzed. Of those who received serum treatment 65 became paralyzed. Was the serum treatment paralyzed?

Ex

In two large populations, there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations.

$$p_1 = \frac{30}{100}$$

$$n_1 = 1200$$

$$p_2 = \frac{25}{100}$$

$$n_2 = 900$$

$$Q_1 = 1 - p_1, Q_2 = 1 - p_2$$

$$= 1 - \frac{3}{10} = \frac{7}{10}$$

$$Q_2 = 1 - \frac{25}{100} = \frac{75}{100}$$

$$n_1 = 1200 \quad n_2 = 900$$

$$p_1 = \frac{3}{10} \quad p_2 = \frac{25}{100} = \frac{1}{4}$$

Test statistic $z = \frac{p_1 - p_2}{S.E.}, \quad S.E. = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$

$$z = \frac{\frac{3}{10} - \frac{25}{100}}{0.02} = 2.55$$

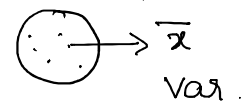
$$S.E. = \sqrt{\frac{\frac{3}{10} \times \frac{7}{10}}{1200} + \frac{\frac{25}{100} \times \frac{75}{100}}{900}}$$

$$H_0: p_1 = p_2$$

$$H_1: p_1 \neq p_2, \quad z = 2.55 > z_{\alpha}, \quad \alpha = 1\%$$

$\Rightarrow$ Reject H_0 .

① $n < 30 \rightarrow$ small samples.



Compare mean $\rightarrow$ Student's t test
Variance $\rightarrow$ F-test.

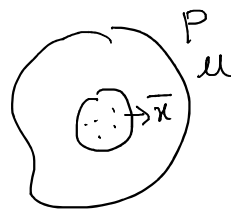
$z_{\alpha} \rightarrow$ degree of freedom.

(Dependent upon no. of samples).

If n is no. of samples, then degree of freedom $n-1$

$$y = mx + c \rightarrow df = 1$$

$$z = ax + by + c \rightarrow df = 2$$



$$H_0: \bar{x} = \mu$$

$$H_1: \bar{x} \neq \mu$$

$$\text{Test statistic } z = \frac{\bar{x} - \mu}{S.E.}, \quad S.E. = \frac{\sigma}{\sqrt{n}}$$

$$|z| < z_{\alpha, df} \Rightarrow \text{Accept } H_0$$

Otherwise Reject H_0 .

Single mean - Large sample

Friday, March 15, 2024 10:30 PM

3. Test of Significance for Single Mean:

To test whether the given sample of size n has been drawn from a population with mean μ .

Null Hypothesis: There is no difference between sample mean $\bar{x}$ and population mean μ .

Test Statistic: $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ where σ - S.D of the population

Note:

1. If the population S.D is not known then the test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \text{ where } s - \text{sample S.D}$$

2. 95% confidence limits for μ are $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$
3. 98% confidence limits for μ are $\bar{x} \pm 2.33 \frac{\sigma}{\sqrt{n}}$
4. 99% confidence limits for μ are $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$

1. A sample of 900 members has a mean of 3.4 cms and a S.D. 2.61 cms. Is the sample from a large population of mean 3.25cm and a S.D. 2.61 cms. If the population is normal and its mean is unknown find the 95% confidence limits of true mean.

population mean: $\mu = 3.25$, $\sigma = 2.61$

Sample mean. $\bar{x} = 3.4$, $n = 900$

$H_0: \bar{x} = \mu$, $H_1: \bar{x} \neq \mu$

Test statistic: $Z = \frac{\bar{x} - \mu}{S.E}$, $S.E = \frac{\sigma}{\sqrt{n}} = \frac{2.61}{\sqrt{900}} = 0.087$

$$Z = \frac{3.4 - 3.25}{S.E} = 1.72$$

$$\alpha = 95\%, \quad Z_{\alpha} = 1.96$$

$|Z| < |Z_{\alpha}| \Rightarrow \text{Accept } H_0.$

95% confidence limits of mean = $\bar{x} \pm Z_{\alpha} S.E$

$$\alpha = 95\%$$

$$= \bar{x} \pm 1.96 * S.E$$

$$= [3.4 - (1.96 \times 0.087),$$

$$3.4 + (1.96 \times 0.087)]$$

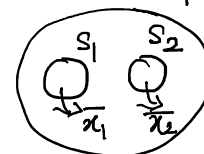
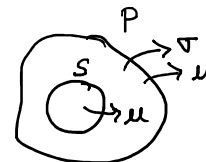
4. Test of Significance for Difference of Mean:

1. To test the significant difference between $\bar{x}_1$ and $\bar{x}_2$

$$\text{Test Statistic: } Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

2. If the samples have been drawn from the same population then $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$\text{Test Statistic: } Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}}$$



$$H_0: \bar{x}_1 = \bar{x}_2$$

$$H_1: \bar{x}_2 \neq \bar{x}_1$$

$$\sigma \quad \sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}$$

Test Statistic:
$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

3. If σ is not known, we can use an estimate of σ^2 given by $\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$

$$H_1: \bar{x}_2 \neq \bar{x}_1$$

$$\frac{\sigma}{\sqrt{n}} = \sqrt{\frac{\sigma^2}{n}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

1. The means of 2 large samples 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D. 2.5 inches.

$$\begin{aligned} S_1: n_1 &= 1000 & S_2: n_2 &= 2000 \\ \bar{x}_1 &= 67.5 & \bar{x}_2 &= 68.0 \\ \sigma_1 &= 2.5 & \sigma_2 &= 2.5 \end{aligned}$$

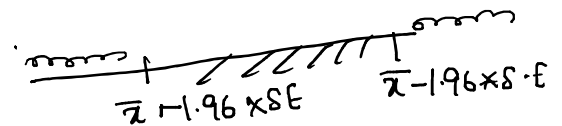
$$H_0: \bar{x}_1 = \bar{x}_2$$

$$H_1: \bar{x}_1 \neq \bar{x}_2$$

$$\begin{aligned} \text{Test statistic } Z &= \frac{\bar{x}_1 - \bar{x}_2}{S.E.}, \quad S.E. = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \\ Z &= \frac{67.5 - 68.0}{S.E.} = \frac{2.5^2}{1000} + \frac{2.5^2}{2000} \\ &= 5.16 \end{aligned}$$

$$Z_{5\%} = 1.96 \Rightarrow |Z| > |Z_{5\%}| \Rightarrow \text{Reject } H_0$$

Confidence $\bar{x} \pm 1.96 S.E$



critical region $(-\infty, \bar{x} - 1.96 * S.E., \bar{x} + 1.96 * S.E., \infty)$

1. A sample of 100 students is taken from a large population. The mean height of the students in this sample is 160. Can it be reasonably regarded that, in the population the mean height is 165cm and S.D is 10cm?

$$\begin{aligned} \text{sample: } n &= 100, \\ \bar{x} &= 160, \mu = 165, \sigma = 10 \end{aligned}$$

$$H_0: \bar{x} = \mu$$

$$H_1: \bar{x} \neq \mu$$

$$S.E. = \frac{\sigma}{\sqrt{n}} = \frac{10}{10} = 1$$

$$\text{Test statistic } Z = \frac{\bar{x} - \mu}{S.E.} = \frac{160 - 165}{1} = -5$$

$$|Z| > |Z_{\alpha}|$$

Reject H_0 .

1. Student's - t test for single mean:

Null hypothesis : $H_0 : \mu = \mu_0$

Alternative hypothesis: $H_1 : \mu \neq \mu_0$

Test statistic: $t = \frac{\bar{x} - \mu}{\frac{S^2}{\sqrt{n}}}$ (S.D is not given directly)

Where $\bar{x} = \frac{\sum x_i}{n}$ = sample mean; μ = population mean

$$S^2 = \frac{\sum (x_j - \bar{x})^2}{n-1}; \quad n = \text{sample size}$$

ii) Test statistic : $t = \frac{\bar{x} - \mu}{\frac{S.D}{\sqrt{n-1}}}$ (S.D is given directly)

Degree of freedom(d.f) : $n-1$

Confidence (or) Fiducial limits for μ

- The 95% confidence limit of the mean of the population is given by

$$\bar{x} \pm t_{0.05} \frac{S}{\sqrt{n}}$$

- The 99% confidence limit of the mean of the population is given by

$$\bar{x} \pm t_{0.01} \frac{S}{\sqrt{n}}$$

- The mean life time of a sample 25 fluorescent light bulbs produced by a company is computed to be 1570 hours with a S.D of 120 hours. The company claims that the average life of the bulbs produced by the company is 1600 hours. Use the level of significance of 5%. Is the claim acceptable?

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

$$\text{sample: } n = 25, \text{ d.f} = n-1 = 24$$

$$S.D = 120$$

$$\mu_0 = 1570$$

$$S.E = \frac{S.D}{\sqrt{n-1}} = \frac{120}{\sqrt{24}}$$

$$\text{Test statistic} = \frac{\mu - \mu_0}{S.E} = \frac{1600 - 1570}{S.E} =$$

$$z = 1.2$$

$$t_{0.05, 24} = 2.06 \Rightarrow |z| < t_{0.05, 24} \Rightarrow H_0 \text{ is accept.}$$

$$\text{confidence interval: } \bar{x} \pm t_{0.05, 24} S.E$$

$$[\bar{x} - t_{0.05, 24} * S.E, \bar{x} + t_{0.05, 24} * S.E]$$

$$\bar{x} = \mu_0$$

$$[1570 - (2.06) * S.E, 1570 + 2.06 * S.E]$$

3. A machinist is making engine parts with axle diameters of 0.700 inch. A random sample of 10 parts shows a mean diameter of 0.742 inch with a S.D of 0.040 inch. Compute the statistic you would use to test whether the work is meeting the specification.

4. A random sample of size 16 valves from a normal population showed a mean of 53 and a sum of squares of deviation from the mean equals to 150. Can this sample be regarded as taken from the population having 56 as mean? Obtain 95% confidence limits of the mean of the population.

$$\text{Sample } n=16, \bar{x}=53, \sum(x-\bar{x})^2=150$$

$$H_0: \mu = 56$$

$$H_1: \mu \neq 56.$$

$$s^2 = \frac{\sum(x-\bar{x})^2}{n-1} = \frac{150}{15} = 10, \quad S.E = \frac{s}{\sqrt{n-1}} = \frac{10}{\sqrt{15}}$$

$$\text{test statistic} = \frac{\bar{x} - 56}{S.E} = \frac{53 - 56}{S.E} = 1.2$$

$$t_{0.05, 15} = 2.13$$

$$|z| < t_{0.05, 15} \Rightarrow \text{Accept } H_0.$$

5. A random sample of 10 boys had the following I.Q's 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100? Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

$n=10,$	x	$x-\bar{x}$	$(x-\bar{x})^2$	$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$
$\bar{x}=97.2$	70			
	120			
	110			
	101			
	88			
	83			
	95			
	98			
	107			
	100			

107

100